

# John Ehrlinger, PhD

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## Research Interests

Applied statistical machine learning research conducted in close collaboration with cardiovascular surgeons and clinicians. Methodological focus spans random forest and ensemble methods, clustering, deep learning, and time series analysis, with emphasis on time-to-event and longitudinal data in cardiovascular outcomes. A sustained focus is the translation of methodological advances into clinical practice through open-source software development and reproducible analytical workflows, with current work centered on open-source implementations of multi-phase hazard analysis methods.

## Professional Experience

*Cleveland Clinic — Heart, Vascular & Thoracic Institute*

Assistant Staff, Lead Data Scientist — Cardiovascular Outcomes, Registries and Research (CORR)    December 2024–Present

Lead a team of data engineers and data scientists supporting cardiovascular outcomes research within CORR. Drive statistical methods research as applied to observational clinical research, advancing rigorous analytic approaches for cardiovascular outcomes data. Champion software engineering best practices and implement process improvements to optimize the departmental research pipeline from data collection through publication.

*Cleveland Clinic Lerner College of Medicine*

Clinical Assistant Professor (Joint Appointment)    June 2025–Present

Serving as faculty mentor and statistical advisor to medical students and trainees conducting clinical research within the Heart, Vascular & Thoracic Institute. Providing guidance in statistical methods, research design, and data analysis for trainee-led research projects.

*Altamira Technologies*

Senior Data Scientist — Technical Lead    March 2023–October 2024

Technical lead of the data science team supporting USAF customers on secure networks. Built a reusable Plotly-based dashboard framework; developed predictive traffic flow models and a computer vision pipeline for vehicle volume estimation at base entry points.

*Microsoft — Azure Global Commercial Industry (AGCI-AI)*

Senior Data and Applied Scientist — Technical Lead    July 2015–April 2023

Technical lead for customer-facing ML/AI engagements across oil & gas (Chevron), aerospace (Rolls Royce), and medical devices (Stryker). Developed Azure ML solution accelerators for predictive maintenance. Presented at Microsoft MLADS conference four times (2016–2020).

## *Cleveland Clinic — Quantitative Health Sciences*

Assistant Staff / Assistant Professor of Medicine

August 2012–July 2015

Applied random survival forests to cardiovascular outcomes research. Initiated development of the ggRandomForests R package. Began migration of institutional SAS-based hazard analysis to open-source implementations.

## *Cleveland Clinic Lerner College of Medicine*

Clinical Assistant Professor (Joint Appointment)

2015

Mentored medical students and fellows in statistical methods and research design as part of joint fellowship programs within the Cardiovascular Outcomes, Registries and Research (CORR) group.

## *Cleveland Clinic — Thoracic and Cardiovascular Surgery*

Lead Systems Analyst: Scientific Programmer

August 1999–August 2012

Supported research computing infrastructure, statistical software development, and HPC operations for the department. Completed doctoral research in Statistics at Case Western Reserve University during this period.

## **Education**

Case Western Reserve University, Ph.D. Statistics

2011

Dissertation: *Regularization: Stagewise Regression and Bagging*. Advisor: Hemant Ishwaran.

Case Western Reserve University, M.S. Mechanical / Aerospace Engineering

1994

Case Western Reserve University, B.S. Mechanical Engineering

1993

## **Honors & Awards**

NASA Graduate Research Fellowship

1992–1994

Cleveland Clinic Innovations Award

2003

## **Publications**

- Robinson, J., Belitsis, G., **Ehrlinger**, J., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCampi, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A. C., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., ... Overman, D. M. (2026). The search for borderline hearts within biventricular repairs of complete atrioventricular septal defects. *The Journal of Thoracic and Cardiovascular Surgery*. <https://doi.org/10.1016/j.jtcvs.2026.05.027>
- Blackstone, E. H., Chang, H. L., Rajeswaran, J., Parides, M. K., Ishwaran, H., Li, L., **Ehrlinger**, J., Gelijns, A. C., Moskowitz, A. J., Argenziano, M., DeRose, J. J., Couderc, J.-P., Balda, D., Dagenais, F., Mack, M. J., Ailawadi, G., Smith, P. K., Acker, M. A., O’Gara, P. T., ... Schering, A. (2019). Biatial maze procedure versus pulmonary vein isolation for atrial fibrillation during mitral valve surgery: New analytical approaches and end points. *The Journal of Thoracic and Cardiovascular Surgery*, 157(1), 234–243.e9. <https://doi.org/10.1016/j.jtcvs.2018.06.093>
- Hurst, T. E., Xanthopoulos, A., **Ehrlinger**, J., Rajeswaran, J., Pande, A., Thuita, L., Smedira, N. G., Moazami, N., Blackstone, E. H., Starling, R. C. (2019). Dynamic prediction of left ventricular assist device pump thrombosis based on lactate dehydrogenase trends. *ESC Heart Failure*, 6(5), 1005–1014. <https://doi.org/10.1002/ehf2.12473>

- Rajeswaran, J., Blackstone, E. H., **Ehrlinger**, J., Li, L., Ishwaran, H., Parides, M. K. (2018). Probability of atrial fibrillation after ablation: Using a parametric nonlinear temporal decomposition mixed effects model. *Statistical Methods in Medical Research*, 27(1), 126–141.
- Wojnarski, C. M., Roselli, E. E., Idrees, J. J., Zhu, Y., Carnes, T. A., Lowry, A. M., Collier, P. H., Griffin, B., **Ehrlinger**, J., Blackstone, E. H., Svensson, L. G., Lytle, B. W. (2018). Machine-learning phenotypic classification of bicuspid aortopathy. *The Journal of Thoracic and Cardiovascular Surgery*, 155(2), 461–469. e4.
- Li, L., Mao, H., Ishwaran, H., Rajeswaran, J., **Ehrlinger**, J., Blackstone, E. H. (2017). Estimating the prevalence of atrial fibrillation from a three class mixture model for repeated diagnoses. *Biometrical Journal*, 59(2), 331–343.
- Pande, A., Li, L., Rajeswaran, J., **Ehrlinger**, J., Kogalur, U. B., Blackstone, E. H., Ishwaran, H. (2017). Boosted multivariate trees for longitudinal data. *Machine Learning*, 106(2), 277–305.
- Alli, O., Rihal, C. S., Suri, R. M., Greason, K. L., Waksman, R., Minha, S., Torguson, R., Pichard, A. D., Mack, M., Svensson, L. G., Rajeswaran, J., Lowry, A. M., **Ehrlinger**, J., Tuzcu, E. M., Thourani, V. H., Makkar, R., Blackstone, E. H., Leon, M. B., Holmes, D. (2016). Learning curves for transfemoral transcatheter aortic valve replacement in the PARTNER-i trial: Technical performance. *Catheterization and Cardiovascular Interventions*, 87(1), 154–162.
- Kapadia, S., Agarwal, S., Miller, D. C., Webb, J. G., Mack, M., Ellis, S., Herrmann, H. C., Pichard, A. D., Tuzcu, E. M., Svensson, L. G., Smith, C. R., Rajeswaran, J., **Ehrlinger**, J., Kodali, S., Makkar, R., Thourani, V. H., Blackstone, E. H., Leon, M. B. (2016). Insights into timing, risk factors, and outcomes of stroke and transient ischemic attack after transcatheter aortic valve replacement in the PARTNER trial (placement of aortic transcatheter valves). *Circulation: Cardiovascular Interventions*, 9(9), e002981.
- Minha, S., Waksman, R., Satler, L. P., Torguson, R., Alli, O., Rihal, C. S., Mack, M., Svensson, L. G., Rajeswaran, J., Blackstone, E. H., Tuzcu, E. M., Thourani, V. H., Makkar, R., **Ehrlinger**, J., Lowry, A. M., Suri, R. M., Greason, K. L., Leon, M. B., Holmes, D. R., Pichard, A. D. (2016). Learning curves for transfemoral transcatheter aortic valve replacement in PARTNER-i trial: Success and safety. *Catheterization and Cardiovascular Interventions*, 87(1), 165–175.
- Raja, S., Rice, T. W., **Ehrlinger**, J., Goldblum, J. R., Rybicki, L. A., Murthy, S. C., Adelstein, D., Videtic, G., McNamara, M. P., Blackstone, E. H. (2016). Importance of residual cancer after induction therapy for esophageal adenocarcinoma. *The Journal of Thoracic and Cardiovascular Surgery*, 152(3), 756–761.
- Suri, R. M., Alli, O., Waksman, R., Rihal, C. S., Satler, L. P., Greason, K. L., Torguson, R., Pichard, A. D., Mack, M., Svensson, L. G., Rajeswaran, J., Lowry, A. M., **Ehrlinger**, J., Mick, S. L., Tuzcu, E. M., Thourani, V. H., Makkar, R., Holmes, D., Leon, M. B., Blackstone, E. H. (2016). Learning curves for transapical transcatheter aortic valve replacement in the PARTNER-i trial: Technical performance, success, and safety. *The Journal of Thoracic and Cardiovascular Surgery*, 152(3), 773–780. e14.
- Smedira, N. G., Blackstone, E. H., **Ehrlinger**, J., Thuita, L., Pierce, C. D., Moazami, N., Starling, R. C. (2015). Current risks of HeartMate II pump thrombosis: Non-parametric analysis of interagency registry for mechanically assisted circulatory support data. *The Journal of Heart and Lung Transplantation*, 34(12), 1527–1534.
- Szeto, W. Y., Svensson, L. G., Rajeswaran, J., **Ehrlinger**, J., Smith, C. R., Mack, M., Miller, D. C., McCarthy, P. M., Bavaria, J. E., Cohn, L. H., Corso, P. J., Guyton, R. A., Thourani, V. H., Lytle, B. W., Williams, M. R., Webb, J. G., Kapadia, S., Tuzcu, E. M., Leon, M. B., Blackstone, E. H. (2015). Appropriate patient selection or healthcare rationing? Lessons from surgical aortic valve replacement in the PARTNER-i trial. *The Journal of Thoracic and Cardiovascular Surgery*, 150(3), 557–568.
- Thourani, V. H., Jensen, H. A., Babaliaros, V., Kodali, S., Rajeswaran, J., **Ehrlinger**, J., Suri, R. M., Don, C. W., Aldea, G., Blackstone, E., Williams, M. R., Makkar, R., Svensson, L., Johnson, A., McCabe, J., Dean, L. S., Johnson, M., Kapadia, S., Cohen, D. J., ... Mackand, M. (2015). PARTNER: Outcomes in nonagenarians undergoing transcatheter aortic valve replacement (TAVR). *Annals of Thoracic Surgery*, 100(3), 785–793.
- Wojnarski, C. M., Roselli, E., Idrees, J., Zhu, Y., Collier, P., Griffin, B., Carnes, T., Lowry, A., **Ehrlinger**, J., Svensson, L., Blackstone, E., Lytle, B. (2015). Bicuspid valve and aneurysm phenotypes: Clinical and pathologic associations. *Journal of the American College of Cardiology*, 65(10).

- Wojnarski, C. M., Svensson, L. G., Roselli, E. E., Idrees, J., Lowry, A. M., **Ehrlinger**, J., Pettersson, G. B., Gillinov, A. M., Johnston, D. R., Soltesz, E. G., Navia, J. L., Hammer, D. F., Griffin, B., Thamilarasan, M., Kalahasti, V., Sabik, J. F., Blackstone, E. H. (2015). Aortic dissection in patients with bicuspid aortic valve– associated aneurysm. *Annals of Thoracic Surgery*, 100(5), 1666–1674.
- Starling, R. C., Moazami, N., Silvestry, S. C., Ewald, G., Rogers, J. G., Milano, C. A., Rame, J. E., Acker, M. A., Blackstone, E. H., **Ehrlinger**, J., Thuita, L., Mountis, M. M., Soltesz, E. G., Lytle, B. W., Smedira, N. G. (2014). Unexpected abrupt increase in left ventricular assist device thrombosis. *New England Journal of Medicine*, 370(1), 33–40. <http://www.nejm.org/doi/full/10.1056/NEJMoa1313385>
- Dalton, J. E., Glance, L. G., Mascha, E. J., **Ehrlinger**, J., Chamoun, N., Sessler, D. I. (2013). Impact of present-on-admission indicators on risk-adjusted hospital mortality measurement. *Anesthesiology*, 118(6), 1298–1306.
- Ehrlinger**, J., Ishwaran, H. (2012). Characterizing  $L_2$ boosting. *The Annals of Statistics*, 40(2), 1074–1101.

## In Revision / Submitted

- Alshneikat, M., Alaraj, R., Awad, A. K., Ramsingh, R., **Ehrlinger**, J., Houghtaling, P. L., Koprivanac, M. J., Kapadia, S., Tong, M. Z., Soltesz, E. G., Smedira, N. G., Roselli, E. E., Wierup, P., Gillinov, A. M., Svensson, L. G., Blackstone, E. H., Pettersson, G. B., Unai, S. G., Bakaeen, F. G. (2026). Coronary artery aneurysms: Evolving trends in surgical management. *The Journal of Thoracic and Cardiovascular Surgery*.
- Barron, J. O., Jain, N., Conner, A., Toth, A. J., **Ehrlinger**, J., Lee, S., Ramji, S., Sudarshan, M., Raymond, D. P., Blackstone, E. H., Murthy, S. C., Raja, S. (2026). Form over function: Development of a morphology-based achalasia classification. *The Journal of Thoracic and Cardiovascular Surgery*.
- Stembal, F., Fang, M. Z., Houghtaling, P., **Ehrlinger**, J., Lieber, E., Kelava, M., Grady, P., Krzelj, K., Indorf, J., Somogyi, D., Troy, A., Lou, X., Kimmaliardjuk, D., Vargo, P., Malas, T., Smedira, N., Tong, M. Z., Soltesz, E., Yun, J., ... Blackstone, E. H. (2026). Is there still room for improvement of cardiopulmonary bypass? Analysis of lactate and other clinical and hemodynamic characteristics. *The Annals of Thoracic Surgery*.

## In Preparation

- Belitsis, G., Robinson, J., **Ehrlinger**, J., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCamp, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., ... Overman, D. (2026). *The search for borderline hearts within complete atrioventricular septal defects*.
- Ehrlinger**, J. (2026). Stability-based clustering for clinicians: A practical guide to SID clustering in cardiovascular research. *The Journal of Thoracic and Cardiovascular Surgery*.
- Ehrlinger**, J., Robinson, J., Belitsis, G., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCamp, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A. C., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., ... Overman, D. M. (2026). *Actual vs. Optimal treatment pathway based on outcomes of complete atrioventricular septal defects using virtual twins*.
- Fang, M. Z., **Ehrlinger**, J., Lieber, E., Krzelj, K., Houghtaling, P., Stembal, F., Kelava, M., Grady, P., Indorf, J., Somogyi, D., Lou, X., Kimmaliardjuk, D., Vargo, P., Malas, T., Smedira, N., Tong, M. Z.-Y., Soltesz, E., Yun, J., Zaki, A., ... Koprivanac, M. (2026). *Is cardiopulmonary bypass providing adequate perfusion? Modeling perioperative lactate levels*.

## Technical Reports

- Ehrlinger**, J. (2016). *ggRandomForests: Exploring Random Forest Survival*. arXiv preprint. <https://arxiv.org/abs/1612.08974>

**Ehrlinger, J.** (2015). *ggRandomForests: Visually Exploring a Random Forest for Regression*. arXiv preprint. <https://arxiv.org/abs/1501.07196>

## Conference Presentations

**Ehrlinger, J.** (2026). *Care and Feeding of Your Biostats Team: Scaling Best Practices in a Large Hybrid SAS/R Team*. Invited talk, R/Medicine Conference. <https://ehrlinger.github.io/CareFeedingBiostats/>

Zamanian, F., **Ehrlinger, J.**, Zhan, C. (2020). *How to interpret model prediction using interpretability tools on Azure Machine Learning*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

**Ehrlinger, J.**, Solanki, P. S. (2019). *Distributed Strategies for Training Deep Learning Models at Scale*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

**Ehrlinger, J.**, Shah, P. (2018). *Automated Machine Learning: A Case Study in Efficiently Expanding Data Science Experimentation for Stryker Surgical IoT Devices*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

**Ehrlinger, J.** (2016). *Are we counting correctly? Model evaluation for time series classification*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

**Ehrlinger, J.** (2014). *Visually Exploring Random Forests with ggRandomForests*. Conference session, useR! 2014, UCLA, Los Angeles, CA.

## Software

### R Packages (CRAN)

*ggRandomForests* — Visual exploration of random forest models via graphical analysis of survival, regression, and classification forests. v3.5.2 on CRAN. <https://CRAN.R-project.org/package=ggRandomForests>

*TemporalHazard* — R port of the C computational core underlying the Cleveland Clinic Hazard SAS module. v1.1.0 on CRAN. <https://CRAN.R-project.org/package=TemporalHazard>

### HVTI R Package Family

Eleven member packages — the nine below, plus *ggRandomForests* and *TemporalHazard* above — resolved from public GitHub repositories and installed, updated and version-checked as a unit by *hvtiR*, a one-command installer and environment diagnostic. <https://github.com/ehrlinger/hvtiR>

*hvtiBoostmtree* — Boosted multivariate trees for longitudinal data; an extended fork of *boostmtree*. <https://github.com/ehrlinger/hvtiBoostmtree>

*hvtiPlotR* — Publication-quality HVTI-standard graphics for reproducible clinical research figures. <https://github.com/ehrlinger/hvtiPlotR>

*hvtiRbootstrap* — Bootstrap model building — fit across many bootstrap replicates and report how often each variable survives selection; an R port of the CORR bootreg, SUMBOOT and cluster SAS macros; in active development. <https://github.com/ehrlinger/hvtiRbootstrap>

*hvtiRdatasets* — Analysis-ready clinical datasets for HVTI CORR studies, with verification against the legacy SAS datasets they replace; in active development. <https://github.com/ehrlinger/hvtiRdatasets>

*hvtiRlifetables* — Age-, sex- and race-matched US reference survival, reproducing the *usmatchd* SAS macro by evaluating a stored three-phase parametric hazard fit rather than interpolating a life table; in active development. <https://github.com/ehrlinger/hvtiRlifetables>

*hvtiRpropensity* — Propensity score estimation, matching, and inverse-probability treatment weighting with standardized balance diagnostics, for cardiac surgery comparative-effectiveness research; in active development. <https://github.com/ehrlinger/hvtiPropensityScores>

*hvtiRtables* — Manuscript-compliant Word tables from gtsummary objects, following HVTI CORR table construction standards, with a JTCVS submission mode. <https://github.com/ehrlinger/hvtiRtables>

*hvtiRtemplates* — Versioned analysis job templates and the analysis-prefix taxonomy the CORR group organizes jobs by, so a study binds to a versioned template rather than to a copy. <https://github.com/ehrlinger/hvtiRtemplates>

*hvtiRutilities* — Utility functions supporting reproducible HVTI research workflows. <https://github.com/ehrlinger/hvtiRutilities>

## **Open-Source Documentation**

*HVTI Recipes* — Working catalog of publication-ready figures, tables, and datasets for clinical outcomes research (Kaplan-Meier, propensity balance, CONSORT, random-forest visualizations), each paired with reproducible code. Quarto book, CC BY 4.0. [https://ehrlinger.github.io/hvti\\_graphics/](https://ehrlinger.github.io/hvti_graphics/)

## **SAS/C Software**

*hazard* — SAS and C implementation of multi-phase hazard analysis for time-to-event decomposition. Includes C source code, SAS macros and modules. (Maintainer) <https://github.com/ehrlinger/hazard>